



American International

University-Bangladesh

(AIUB)

**Department of Computer Science
Faculty of Science & Technology (FST)**

***AI-Powered Academic Stress Management Application For
University Students***

A Software Engineering Project Submitted

By

Semester: Summer_23_24		Section:F	Group Number: 5	
SN	Student Name	Student ID	Contribution (CO3+CO4)	Individual Marks
1	MD HASIBUL ISLAM SHANTO	22-48880-3		
2	MAHIYA AKTHER	22-47359-2		
3	TAH YEAMIN SARKER	22-46781-1		
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The project will be Evaluated for the following Course Outcomes

CO3: <i>Select</i> appropriate software engineering models, project management roles and their associated skills for the complex software engineering project and evaluate the sustainability of developed software, taking into consideration the societal and environmental aspects	Total Marks	
	Appropriate Process Model Selection and Argumentation with Evidence	[5 Marks]
	Evidence of Argumentation regarding process model selection	[5Marks]
	Analysis the impact of societal, health, safety, legal and cultural issues	[5Marks]
Submission, Defense, Completeness, Spelling, grammar and Organization of the Project report	[5Marks]	
CO4: <i>Develop</i> project management plan to manage software engineering projects following the principles of engineering management and economic decision process	Total Marks	
	Develop the project plan, its components of the proposed software products	[5Marks]
	Identify all the activities/tasks related to project management and categorize them within the WBS structure. Perform detailed effort estimation correspond with the WBS and schedule the activities with resources	[5Marks]
Identify all the potential risks in your project and prioritize them to overcome these risk factors.	[5Marks]	

Description of Student's Contribution in the Project work

Student Name: MD HASIBUL ISLAM SHANTO Student ID: 22-48880-3 Contribution in Percentage (%): 20 <u>Contribution in the Project:</u> • Researched and drafted the "Background to the Problem." • Developed and described the comprehensive solution features. • Defined roles and responsibilities for various project roles. • Created the use case diagram. _____ Signature of the Student
Student Name: MAHIYA AKTHER Student ID: 22-47359-2 Contribution in Percentage (%): 20 <u>Contribution in the Project:</u> • Developed the activity diagram. • Drafted the functional requirements for "Sign Up," "Login," and "Forgot Password." • Developed the functional requirements for "Chatbot for Empathetic Conversation and Support" and "Guided Meditation and Stress Management Tasks." _____ Signature of the Student
Student Name: TAH YEAMIN SARKER Student ID: 22-46781-1 Contribution in Percentage (%): 20 <u>Contribution in the Project:</u> • Explained the advantages of using the Scrum process model. • Drafted the functional requirements for "Secure Payment Processing," "Subscription Plans," and "Free and Premium Plan Management." _____ Signature of the Student
Student Name: NAFIS HASAN KHAN Student ID: 22-46688-1 Contribution in Percentage (%): 20 <u>Contribution in the Project:</u> • Created the class diagram. • Conducted the sustainability evaluation. • Drafted the functional requirements for "User Account Management" and "Data Privacy and Security."

Signature of the Student

Student Name: MD. HASAN

Student ID: 22-46548-1

Contribution in Percentage (%): 20

Contribution in the Project:

- Created the sequence diagram.
- Analyzed societal, health, safety, legal, and cultural impacts.
- Drafted the functional requirements for "Therapy Session Booking and Management" and "Community Forum."

Signature of the Student

1. PROJECT PROPOSAL

1.1 Background to the Problem

Academic stress is a pervasive issue affecting students globally. With increasing academic demands, students face immense pressure to perform well, manage their time efficiently, and balance various aspects of their lives. This stress can lead to severe mental health issues, including anxiety, depression, and burnout. The COVID-19 pandemic has further exacerbated these issues, making remote learning a challenging experience for many.

The root cause of this problem is multifaceted:

- **High Expectations:** Academic institutions and societal expectations place significant pressure on students to excel.
- **Lack of Support:** There is often a lack of accessible support systems for students dealing with academic stress.
- **Time Management:** Students struggle to balance coursework, extracurricular activities, and personal life.
- **Mental Health Stigma:** Many students hesitate to seek help due to the stigma associated with mental health issues.

This problem is crucial to address because prolonged academic stress can lead to serious health consequences and negatively impact students' future prospects.

1.2 Solution to the Problem

The objective of this project is to develop a comprehensive chatbot application that provides empathetic conversation, guided meditation, stress management tasks, and access to professional support to help students manage academic stress effectively.

To address the identified problem, the proposed solution includes the following key features:

- **Empathetic Chatbot:** Using Natural Language Processing (NLP), the chatbot will engage in empathetic conversations with students, providing immediate support and guidance.
- **Guided Meditation and Stress Management:** The app will offer guided meditation sessions, breathing exercises, and focus training modules to help students manage stress.
- **Therapy Session Booking:** Integration with professional therapists will allow students to book therapy sessions directly through the app.
- **Community Forum:** A platform for students to share their experiences and support each other, either anonymously or non-anonymously.
- **Personalized AI-Driven Recommendations:** Based on user data, the app will provide personalized strategies and recommendations for managing stress and improving academic performance.
- **Progress Tracking and Goal setting:** Tools to set goals, track progress, and visualize improvements over time.

This solution is particularly appropriate because it leverages state-of-the-art technology to provide scalable and accessible mental health support. The integration of AI for personalized recommendations ensures that students receive relevant and effective strategies. Additionally, the community forum and professional support features create a holistic support system that addresses various aspects of academic stress.

Basic Functionalities of the Proposed Solution: The Academic Stress Support Chatbot leverages state-of-the-art technology to deliver a comprehensive solution to the problem of academic stress among students. Here are the key functionalities and their impacts:

- **Empathetic Chatbot:** Utilizing Natural Language Processing (NLP), the chatbot engages in empathetic conversations with students, offering immediate support and guidance. This feature leverages advanced AI to understand and respond to user emotions, providing a personalized and comforting experience.
- **Guided Meditation and Stress Management:** The app offers guided meditation sessions, breathing exercises, and focus training modules to help students manage stress. These features use evidence-based techniques to promote mental well-being and are personalized based on user engagement and feedback.
- **Therapy Session Booking:** Integration with professional therapists allows students to book therapy sessions directly through the app. This feature includes secure payment gateways and provides access to licensed professionals, ensuring that students receive timely and appropriate mental health care.
- **Community Forum:** A platform for students to share their experiences and support each other, either anonymously or non-anonymously. This forum fosters a sense of community and peer support, which is crucial for reducing feelings of isolation and promoting mental health.
- **Personalized AI-Driven Recommendations:** Based on user data, the app provides personalized strategies and recommendations for managing stress and improving academic performance. Machine learning algorithms continuously update these recommendations to ensure they remain relevant and effective.
- **Progress Tracking and Goal-Setting:** Tools to set goals, track progress, and visualize improvements over time. These features help students stay motivated and see the tangible benefits of their stress management efforts.
- **User Account Management:** Secure sign-up, login, and password recovery processes ensure that user data is protected and easily accessible.
- **Cross-Platform Support:** Availability on Windows, Mac, iOS, and Android devices ensures that students can access support wherever they are.
- **Data Privacy and Security:** Compliance with data privacy regulations (e.g., GDPR, CCPA) ensures that user data is encrypted and handled securely.
- **Push Notifications:** Reminders, updates, and motivational messages keep users engaged and informed.

Target Group of Users: The primary target group includes students at various educational levels who are experiencing academic stress. Secondary users include educational institutions looking to provide better mental health support for their students.

Analysis of Societal, Health, Safety, Legal, and Cultural Impact

The application reduces stigma around mental health by providing a private, non-judgmental space for students to seek help. It fosters a supportive community that encourages open discussions about mental health, helping to break down cultural barriers. Immediate access to mental health support and guided stress management techniques can significantly reduce symptoms of anxiety and depression. Personalized recommendations ensure that the support provided is relevant and effective, promoting overall mental well-being.

The application includes crisis intervention features that recognize distress signals and direct users to emergency resources, ensuring that students receive timely help in critical situations. Compliance with data privacy regulations protects user data, building trust and ensuring that the app operates within legal frameworks. Secure transactions and data handling practices mitigate risks related to user privacy and security.

Moreover, multi-language support and the option for anonymous interaction respect cultural differences and reduce barriers to seeking help. The app's inclusive design ensures that it is accessible to a diverse student population. By combining these aspects, the application offers a holistic approach to mental health support, addressing the immediate needs of students while promoting long-term societal well-being.

The Academic Stress Support Application stands out by integrating advanced AI technologies with comprehensive mental health resources. The empathetic chatbot and AI-driven recommendations provide personalized, real-time support, while the integration of professional therapy booking and community forums creates a robust support network. This innovative solution not only addresses academic stress but also enhances overall mental health, making a significant positive impact on the lives of students.

This project will contribute to the development of innovative mental health support tools tailored for the academic environment. The use of AI for personalized recommendations and the integration of a community support system are expected to produce significant results in improving student well-being. Previous studies have shown that AI-driven mental health applications can improve mental health outcomes by providing timely and personalized support [1]. Additionally, online community support systems have proven effective in providing emotional and informational support, which is crucial for mental health [2].

Existing studies on academic stress have highlighted the importance of accessible mental health support and the potential of digital tools to provide such support. Research indicates that a significant number of college students suffer from high levels of stress, anxiety, and depression, necessitating accessible mental health resources [3]. Digital mental health interventions have been shown to effectively reduce symptoms of depression and anxiety among university students [4]. Furthermore, chatbots using natural language processing (NLP) can offer significant support by providing real-time, empathetic responses to users' concerns [5].

Current solutions include standalone meditation apps, therapy booking platforms, and academic support services. Examples include:

- **Woebot:** An AI chatbot offering cognitive-behavioral therapy (CBT) techniques through chat-based interactions [6]. While effective in delivering CBT, it lacks integration with professional therapy booking and community support features.
- **Headspace:** A popular meditation app providing guided meditation and mindfulness exercises [7]. It does not, however, offer personalized academic stress management or professional support booking.
- **BetterHelp:** An online platform for booking therapy sessions with licensed professionals [8]. It offers extensive therapy options but does not integrate community support or AI-driven stress management tools.
- **TalkSpace:** Provides therapy via text, voice, and video with licensed therapists [9]. Similar to BetterHelp, it lacks community forums and AI-based personalized recommendations.

Our app aims to fill these gaps by integrating empathetic chatbot features to provide immediate emotional support, offering guided meditation and stress management tasks tailored to academic stress, enabling professional therapy session booking within the app, providing a community forum for peer support and shared experiences, and utilizing AI to deliver personalized recommendations for stress management and academic improvement. By combining these functionalities, our solution provides a comprehensive approach to managing academic stress, addressing the limitations of existing applications, and offering a more effective and integrated support system for students.

2. SOFTWARE DEVELOPMENT LIFE CYCLE

2.1 Process Model

The proposed software is a comprehensive mental health application aimed at managing academic stress among students. It incorporates multiple features to provide holistic support, including an empathetic chatbot, guided meditation, therapy session booking, a community forum, personalized AI-driven recommendations, and progress tracking.

The software will be deployed on multiple platforms, including Windows, Mac, iOS, and Android, to ensure accessibility. It will utilize cloud services for data storage, processing, and AI functionalities, ensuring scalability and reliability. The target users are students from diverse backgrounds and educational institutions, necessitating a user-centric design with considerations for accessibility, privacy, and cultural sensitivity.

Advantages of Using Scrum:

- **Flexibility and Adaptability:** The software includes a range of features that may evolve based on user feedback and emerging needs. Scrum handles changing requirements effectively through iterative sprints, allowing for regular reassessment and adaptation.
- **User and Client Involvement:** Regular interaction with stakeholders through sprint reviews and client demos ensures that the development process remains aligned with user needs and expectations. This continuous feedback loop is critical for developing user-centric features.

- **Risk Management:** Scrum allows for early identification and mitigation of risks. Each sprint provides an opportunity to review progress, address issues, and adjust plans as needed, reducing the likelihood of significant issues arising late in the project.
- **Team Collaboration:** Scrum encourages collaboration among cross-functional teams, which is vital for integrating various components such as AI-driven recommendations, mental health resources, and user interfaces. Daily stand-up meetings ensure alignment and quick resolution of blockers.
- **Incremental Delivery and Continuous Improvement:** By delivering functional increments at the end of each sprint, Scrum ensures that each feature is developed, tested, and validated in phases. This enables early testing and validation, making the development process more efficient. Scrum includes regular retrospectives, allowing the team to reflect on their processes and make continuous improvements, enhancing overall project efficiency and effectiveness.

Justification for Choosing Scrum:

- **Successful Projects:** Numerous case studies in the tech industry have demonstrated the success of Scrum in managing complex software projects with evolving requirements and diverse features.
- **Wide Adoption:** Scrum is widely adopted in the software development industry, particularly for projects that require flexibility and iterative progress. Its popularity and proven track record provide strong evidence for its effectiveness.
- **Improved Productivity:** Research studies have shown that Scrum improves team productivity, collaboration, and the ability to handle changes in project scope, making it ideal for developing a dynamic and user-focused application.

Evaluation of Sustainability:

The sustainability evaluation of the proposed software encompasses both societal and environmental dimensions. The application aims to reduce academic stress and improve mental health among students. By providing accessible tools for stress management, the application can significantly benefit the student community. Features like community forums promote peer support, fostering a sense of belonging and reducing feelings of isolation.

The software is designed to be accessible to a diverse student population, including those with disabilities, and supports multiple languages, enhancing its reach and impact. It provides a private, non-judgmental space for students to seek help, reducing the stigma associated with mental health issues.

Implementing efficient coding practices to minimize energy consumption on user devices includes optimizing the application to be lightweight and efficient. Using servers powered by renewable energy sources to host the application reduces its environmental impact. Choosing energy-efficient data centers with low carbon footprints and implementing resource optimization techniques such as load balancing and efficient storage use help minimize the digital carbon footprint.

Encouraging remote work for the development team reduces the environmental impact associated with commuting and office space usage. Adopting paperless processes and using digital tools for collaboration and documentation further reduce the environmental footprint of the project.

By adopting these sustainable practices, the developed software will have a positive societal impact while minimizing its environmental footprint. Continuous evaluation and transparent reporting of these sustainability metrics will ensure the long-term sustainability of the application.

2.2 Project Role Identification and Responsibilities

Product Owner

- Responsibilities: Defines the product backlog, prioritizes features, and ensures the development team delivers value to the business. Represents stakeholders.
- Skills: Requirement analysis, stakeholder management, prioritization, decision-making.

Project Manager

- Responsibilities: Oversees the entire project, ensures timelines, budget, and quality standards are met. Coordinates between teams and manages resources.
- Skills: Leadership, communication, risk management, budgeting, scheduling.

Scrum Master

Responsibilities: Facilitates Scrum practices, removes impediments, and ensures smooth communication between the team and the Product Owner. Coaches the team on Agile principles.

Skills: Facilitation, conflict resolution, coaching, Agile principles.

Development Team

- Responsibilities: Builds the application, including coding, designing, testing, and integrating various features.
- Skills: Software development, testing, design, problem-solving, collaboration.

Quality Assurance (QA) Engineer

- Responsibilities: Ensures the product meets quality standards through rigorous testing and validation.
- Skills: Test planning, automation, defect tracking, attention to detail.

UX/UI Designer

- Responsibilities: Designs the user interface and experience, ensuring the application is user-friendly and accessible.
- Skills: User research, prototyping, visual design, usability testing.

AI Specialist

- Responsibilities: Develops and refines the AI-driven features, ensuring robustness and effectiveness.

- Skills: Machine learning, natural language processing, data modeling, software engineering.

Mental Health Professional

- Responsibilities: Ensures the application's features, such as the chatbot and guided meditations, are effective and appropriate.
- Skills: Clinical expertise, empathy, counseling, crisis intervention.

Data Analyst

- Responsibilities: Analyzes user data to provide insights for personalized recommendations and improvements.
- Skills: Data analysis, statistics, machine learning, data visualization.

DevOps Engineer

- Responsibilities: Manages the deployment, monitoring, and maintenance of the application infrastructure. Ensures continuous integration and continuous deployment (CI/CD) processes are implemented. Automates development, testing, and deployment processes.
- Skills: Automation tools, cloud platforms (AWS, Azure, Google Cloud), CI/CD pipelines, containerization (Docker, Kubernetes), scripting (Python, Bash).

Security Engineer

- Responsibilities: Ensures the application is secure from threats and vulnerabilities. Implements security best practices, conducts security testing, and monitors for security breaches. Educates the team on security protocols and practices.
- Skills: Cybersecurity, penetration testing, security tools (e.g., OWASP), encryption, network security, incident response.

Business Analyst

- Responsibilities: Analyzes business processes, identifies requirements, and ensures that the developed software meets business objectives. Works closely with stakeholders to gather and document requirements.
- Skills: Business process modeling, requirement gathering, documentation, stakeholder communication, analytical thinking.

Compliance Officer

- Responsibilities: Ensures that the application complies with relevant laws and regulations, particularly concerning data privacy and security. Monitors compliance and conducts regular audits.
- Skills: Knowledge of regulations (e.g., GDPR, CCPA), auditing, risk assessment, legal understanding, attention to detail.

Content Manager

- Responsibilities: Manages the content within the application, including guided meditations, articles, and other resources. Ensures content is relevant, up-to-date, and culturally sensitive.

- Skills: Content creation, content management systems (CMS), SEO, editorial skills, cultural sensitivity.

Customer Support Specialist

- **Responsibilities:** Provides support to users, addressing issues and collecting feedback. Helps users navigate the application and resolve any technical problems they encounter.
- Skills: Customer service, communication, problem-solving, technical support.

Training Specialist

- Responsibilities: Develops training materials and conducts training sessions for users and staff. Ensures that all users are adequately trained to use the application effectively.
- Skills: Instructional design, training delivery, communication, technical writing, public speaking.

3. REQUIREMENT ANALYSIS

3.1 Functional Requirements

1. Sign Up

Functional Requirements:

- 1.1 The software shall allow new users to create an account by providing necessary details such as student ID, email, username, and password.
- 1.2 The system shall verify the provided email address by sending a verification link.
- 1.3 Upon clicking the verification link, the user account will be activated, and the user can proceed to log in.
- 1.4 The software shall ensure that the username, student ID, and email are unique and not already taken by another user.
- 1.5 The password must meet security requirements, including a minimum length and the inclusion of special characters.
- 1.6 The system shall provide feedback to the user if the provided information does not meet the requirements (e.g., username taken, existing email, existing student ID, weak password).
- 1.7 The software shall offer the option to sign up using social media accounts (e.g., Google, Facebook) for convenience.

Priority Level: High

Precondition: Users have access to a valid email address or a social media account for registration.

2. Login

Functional Requirements:

- 2.1 The software shall allow existing users to log in using their email and password.
- 2.2 The system shall provide feedback if the login credentials are incorrect.
- 2.3 The software shall offer the option to log in using social media accounts (e.g., Google, Facebook).
- 2.4 The system shall support multi-factor authentication for enhanced security.

2.5 The software shall provide an option for users to stay logged in on their devices.

Priority Level: High

Precondition: Users have an active account.

3. Forgot Password

Functional Requirements:

3.1 The software shall provide a "Forgot Password" option on the login page.

3.2 The system shall prompt the user to enter their registered email address.

3.3 The software shall send a password reset link to the provided email address.

3.4 Upon clicking the reset link, users shall be able to create a new password.

3.5 The new password must meet security requirements, including a minimum length and the inclusion of special characters.

Priority Level: High

Precondition: Users have a registered account and access to their email.

4. Account Security Features

Functional Requirements:

4.1 The software shall offer password recovery and forgot password functionality.

4.2 The system shall support multi-factor authentication for enhanced security.

4.3 The software shall notify users of any suspicious account activity.

4.4 The system shall provide security tips and best practices for users.

4.5 The software shall enforce account lockout after multiple failed login attempts.

Priority Level: High

Precondition: Users have an active account.

5. Secure Payment Processing

Functional Requirements:

5.1 The software shall implement secure payment processing for premium features and services, complying with industry security standards such as PCI DSS.

5.2 The system shall encrypt sensitive payment information during transmission using TLS/SSL protocols and store payment information securely.

5.3 The software shall support fraud detection and prevention mechanisms, including monitoring and alerting for suspicious activity.

5.4 The system shall provide users with the ability to securely save and manage their payment methods.

5.5 The software shall provide users with transaction history and receipts for all payments made through the app.

Priority Level: High

Precondition: Users have access to the application and valid payment methods.

6. Subscription Plans

Functional Requirements:

6.1 The software shall offer flexible subscription plans, including monthly and yearly options, for full access to premium features.

6.2 The system shall provide automatic renewal options for subscription plans, allowing users to enable or disable this feature.

6.3 The software shall allow users to easily cancel their subscription at any time through the user account settings.

6.4 The system shall notify users via email and in-app notifications before the automatic renewal of their subscription, including details of the renewal date and amount.

6.5 The software shall handle proration for users upgrading or downgrading their subscription plans mid-cycle.

6.6 The system shall support promotional codes and discounts for subscription plans.

Priority Level: High

Precondition: Users have an active account and valid payment methods.

7. Free and Premium Plan Management

Functional Requirements:

7.1 The software shall provide a free plan with access to a limited set of basic features.

7.2 The system shall allow users to upgrade from the free plan to premium plans seamlessly.

7.3 The software shall clearly differentiate between features available on the free plan and those available on premium plans using UI elements.

7.4 The system shall notify users of the benefits of upgrading to a premium plan during their interaction with the app.

7.5 The software shall track the user's usage of free plan features and recommend upgrades based on usage patterns.

7.6 The software shall ensure that all premium features are accessible exclusively to users with active subscription plans.

7.7 The system shall verify the subscription status of users before granting access to premium features.

7.8 The software shall provide notifications to users when their subscription is about to expire, including options to renew or upgrade.

7.9 The system shall restrict access to premium features immediately if the subscription expires or is canceled.

7.10 The software shall provide a grace period for expired subscriptions, during which users can renew without losing access to premium features.

Priority Level: High

Precondition: Users have an active account.

8. Payment Gateway Integration

Functional Requirements:

8.1 The software shall integrate with multiple payment gateways (e.g., credit/debit cards, mobile wallets) to support secure and flexible transactions.

8.2 The system shall support the setup and management of multiple payment methods within a user account, allowing users to add, remove, and update payment details.

8.3 The software shall ensure that all transactions are processed securely, following best practices for data protection, including CVV verification and 3D Secure.

8.4 The system shall handle payment failures gracefully, providing users with clear error messages and options to retry or use a different payment method.

8.5 The software shall support refunds and chargebacks through the integrated payment gateways, with clear policies and processes for users to follow.

8.6 The system shall provide detailed transaction logs and reports for administrative purposes, including payment statuses, timestamps, and user details.

Priority Level: High

Precondition: Users have an active account and valid payment methods.

9. Chatbot for Empathetic Conversation and Support

Functional Requirements:

9.1 The software shall provide a chatbot capable of empathetic conversations to support users experiencing academic stress.

9.2 The chatbot shall offer predefined responses and allow for dynamic conversation flow based on user inputs.

9.3 The system shall use natural language processing (NLP) to understand and respond to user queries accurately.

9.4 The chatbot shall escalate serious issues to human support when necessary.

9.5 The system shall log interactions for continuous improvement of responses and support.

Priority Level: High

Precondition: Users have access to the application and have completed onboarding.

10. Guided Meditation and Stress Management Tasks

Functional Requirements:

10.1 The software shall provide a variety of guided meditation sessions for different durations.

10.2 The system shall include breathing exercises with visual and auditory guides.

10.3 The software shall offer focus training tasks, such as concentration exercises and mindfulness activities.

10.4 Users shall be able to choose the type of meditation or stress management task based on their current needs.

10.5 The software shall track user participation in meditation and stress management tasks.

10.6 The system shall provide progress reports on user engagement.

10.7 The software shall offer personalized recommendations for meditation and stress management tasks.

10.8 The system shall support offline access for guided exercises and stress management techniques.

10.9 The software shall provide notifications and reminders for scheduled sessions.

10.10 The system shall allow users to set goals for regular practice and track progress.

Priority Level: High

Precondition: Users have access to the application and a stable internet connection for initial content download.

11. Therapy Session Booking and Management

Functional Requirements:

11.1 The software shall allow users to book therapy sessions with experts through the app.

11.2 The system shall integrate with payment gateways to process payments for these sessions.

11.3 The software shall provide options to view, reschedule, and cancel appointments.

11.4 The system shall send reminders for upcoming sessions via email or push notifications.

11.5 The software shall allow users to view the profiles and specialties of available therapists.

11.6 The software shall provide a calendar view of all booked sessions.

11.7 The system shall allow users to provide feedback on their sessions.

11.8 The software shall handle booking conflicts and suggest alternative times.

Priority Level: High

Precondition: Users have an active account, have set up payment details, and have booked therapy sessions.

12. Community Forum

Functional Requirements:

12.1 The software shall provide a forum for users to share their problems anonymously or non-anonymously.

12.2 The system shall support posting, commenting, and liking functionalities.

12.3 The software shall include moderation features to ensure a safe and supportive environment.

12.4 The system shall allow users to report inappropriate content.

12.5 The software shall categorize forum posts by topics for easy navigation.

Priority Level: High

Precondition: Users have an active account and are logged in.

13. Personalized AI-Driven Recommendations

Functional Requirements:

13.1 The software shall analyze user data to provide personalized recommendations for managing academic stress and improving grades.

13.2 The system shall use machine learning algorithms to update recommendations based on user behavior and feedback.

13.3 The software shall display recommendations in the user dashboard.

13.4 The system shall provide explanations for the recommendations given to users.

13.5 The software shall allow users to give feedback on the recommendations received.

Priority Level: High

Precondition: Users have an active account and are logged in.

14. Academic and Career Guidance

Functional Requirements:

14.1 The software shall provide access to academic and career guidance from experts.

14.2 The system shall allow users to book sessions with academic and career advisors.

14.3 The software shall integrate with payment gateways for any paid guidance sessions.

14.4 The system shall provide profiles of available advisors, including their expertise and availability.

14.5 The software shall allow users to submit questions and receive written advice from experts.

Priority Level: High

Precondition: Users have an active account and have set up payment details.

15. Progress Tracking and Goal-Setting

Functional Requirements:

15.1 The software shall allow users to set goals and track their progress.

15.2 The system shall aggregate data to show overall progress in stress management and academic performance.

15.3 The software shall provide visualizations of progress in the user dashboard.

15.4 The system shall send reminders and motivational messages to encourage users to stay on track with their goals.

15.5 The software shall allow users to adjust their goals and update progress logs.

Priority Level: High

Precondition: Users have an active account and are logged in.

16. Cross-Platform Support

Functional Requirements:

16.1 The software shall be compatible with Windows, Mac, iOS, and Android.

16.2 The system shall provide a consistent user experience across all platforms.

16.3 The software shall synchronize user data across all devices.

16.4 The system shall support installation and updates through official app stores and websites.

16.5 The software shall optimize performance for each platform to ensure smooth operation.

Priority Level: High

Precondition: Users have an active account and use multiple devices.

17. User Data Synchronization

Functional Requirements:

17.1 The software shall synchronize user data across all devices in real-time.

17.2 The system shall handle data conflicts and provide resolution mechanisms.

17.3 The software shall allow users to manually trigger data synchronization.

17.4 The system shall ensure data integrity during synchronization processes.

17.5 The software shall notify users of successful synchronization or any issues encountered.

Priority Level: High

Precondition: Users have an active account and use multiple devices.

18. User Account Management

Functional Requirements:

18.1 The software shall allow users to create and manage their accounts.

18.2 The system shall support profile updates, including username, email, and password changes.

18.3 The software shall provide options for users to manage privacy settings.

18.4 The system shall allow users to delete their accounts and data permanently.

18.5 The software shall notify users of any changes made to their accounts.

Priority Level: High

Precondition: Users have an active account.

19. Resource Library

Functional Requirements:

19.1 The software shall provide a library of articles, videos, and other materials on stress management and mental health.

19.2 The system shall allow users to search and filter resources by category.

19.3 The software shall recommend resources based on user interests and behavior.

19.4 The system shall regularly update the resource library with new content.

19.5 The software shall provide options for users to save and bookmark resources.

Priority Level: Should Have

Precondition: Users have an active account.

20. Crisis Hotline Access

Functional Requirements:

20.1 The software shall provide immediate access to crisis hotlines or emergency support.

20.2 The system shall include a prominent emergency button in the user interface.

20.3 The software shall list contact details for various crisis support services.

20.4 The system shall ensure hotline information is up-to-date and accessible.

20.5 The software shall allow users to directly call or message crisis hotlines from the app.

Priority Level: Should Have

Precondition: Users have an active account and require emergency support.

21. Peer Mentorship Program

Functional Requirements:

21.1 The software shall facilitate a peer mentorship program connecting users with senior students or trained mentors.

21.2 The system shall support mentor-mentee matching based on preferences and availability.

21.3 The software shall allow users to schedule and track mentorship sessions.

21.4 The system shall provide resources and guidelines for mentors and mentees.

21.5 The software shall include feedback mechanisms to evaluate mentorship effectiveness.

Priority Level: Should Have

Precondition: Users have an active account and opt-in to the mentorship program.

22. Personalized User Dashboard

Functional Requirements:

22.1 The software shall provide a personalized dashboard showing user progress, upcoming tasks, and recommendations.

22.2 The system shall update the dashboard in real-time based on user activity.

22.3 The software shall allow users to customize the dashboard layout and content.

22.4 The system shall provide visualizations and summaries of user data.

22.5 The software shall highlight important tasks and deadlines in the dashboard.

Priority Level: Should Have

Precondition: Users have an active account.

23. University Portal Integration

Functional Requirements:

23.1 The software shall integrate with university portals for seamless access to academic information.

23.2 The system shall synchronize academic data from the university portal with the app.

23.3 The software shall allow users to view academic schedules, grades, and announcements.

23.4 The system shall ensure data security and compliance with university policies.

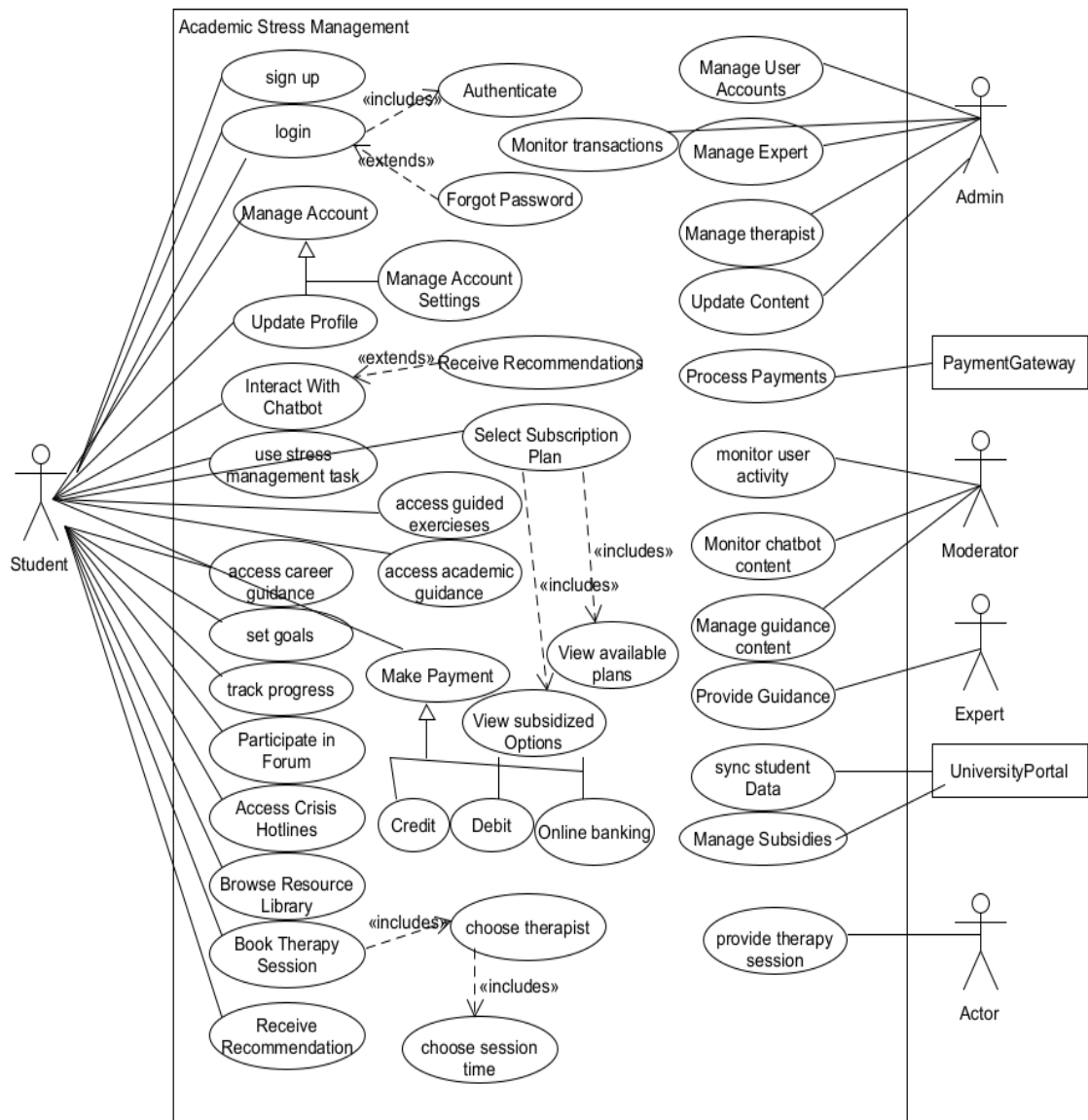
23.5 The software shall provide notifications for important academic updates.

Priority Level: Could Have

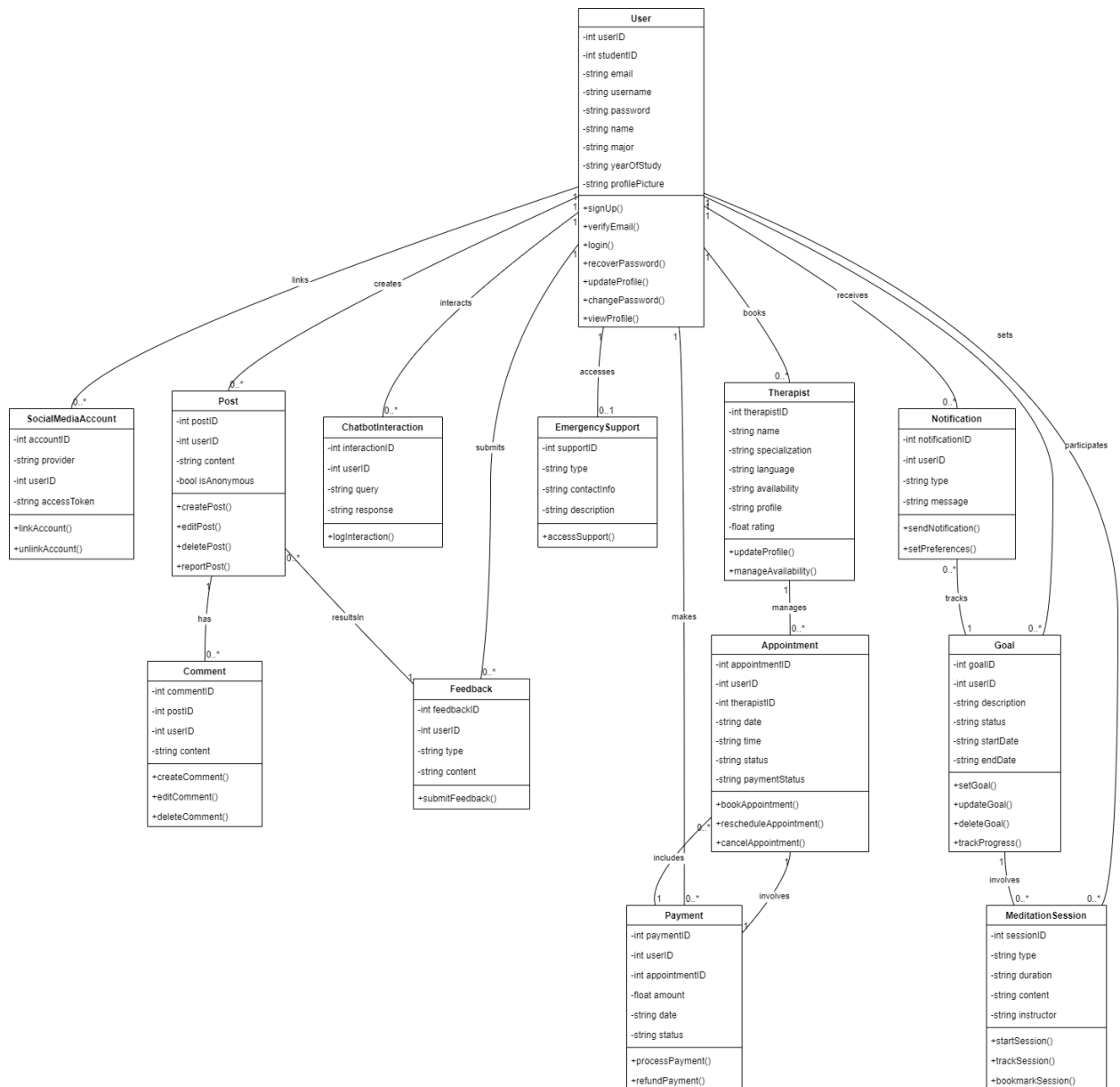
Precondition: Users have an active account and university portal access.

4. SYSTEM DESIGN SPECIFICATION

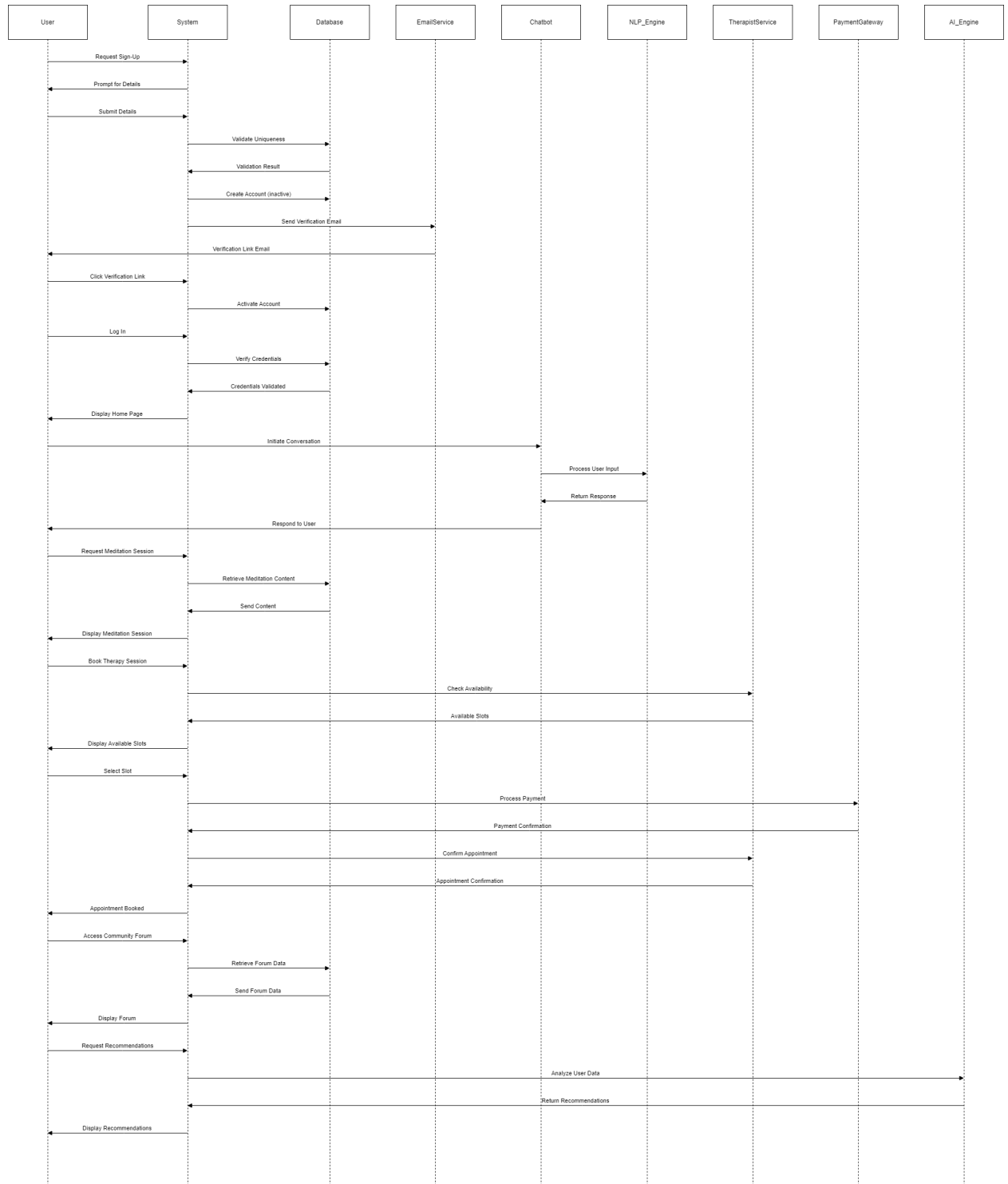
- Use Case Diagram



- Class Diagram



- Sequence Diagram



- Activity Diagram

References

- [1] K. K. Fitzpatrick, A. Darcy, and M. Vierhile, "Delivering cognitive behavior therapy to young adults with symptoms of depression and anxiety using a fully automated conversational agent (Woebot): A randomized controlled trial," *JMIR Mental Health*, vol. 4, no. 2, p. e19, 2017.
- [2] L. A. Ellis, P. Collin, P. J. Hurley, T. A. Davenport, J. M. Burns, and I. B. Hickie, "Young men, mental health, and technology: Implications for service design and delivery in the digital age," *Journal of Medical Internet Research*, vol. 16, no. 6, p. e143, 2014.
- [3] R. Beiter, R. Nash, M. McCrady, D. Rhoades, M. Linscomb, M. Clarahan, and S. Sammut, "The prevalence and correlates of depression, anxiety, and stress in a sample of college students," *Journal of Affective Disorders*, vol. 173, pp. 90-96, 2015.
- [4] D. D. Ebert, P. Mortier, F. Kaehele, R. Bruffaerts, H. Baumeister, R. P. Auerbach, and R. C. Kessler, "Barriers of mental health treatment utilization among first-year college students: First cross-national results from the WHO World Mental Health International College Student Initiative," *International Journal of Methods in Psychiatric Research*, vol. 27, no. 2, p. e1725, 2018.
- [5] B. Inkster, S. Sarda, and V. Subramanian, "An empathy-driven, conversational artificial intelligence agent (Wysa) for digital mental well-being: Real-world data evaluation mixed-methods study," *JMIR mHealth and uHealth*, vol. 6, no. 11, p. e12106, 2018.
- [6] Woebot Labs, "Woebot: Your charming robot friend who is here for you, 24/7," [Online]. Available: <https://woebot.io/>
- [7] Headspace Inc., "Headspace: Meditation and mindfulness made simple," [Online]. Available: <https://www.headspace.com/>
- [8] BetterHelp, "BetterHelp: Online counseling & therapy," [Online]. Available: <https://www.betterhelp.com/>
- [9] Talkspace, "Talkspace: Online therapy with a licensed therapist," [Online]. Available: <https://www.talkspace.com/>